



Oxidation kinetics of methyl crotonate: A comprehensive modeling and experimental study

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ABSTRACT

The current study explores the combustion behavior of methyl crotonate ($\text{CH}_3\text{CH}=\text{CHC}(=\text{O})\text{OCH}_3$), which is a short ester representative of large unsaturated methyl esters. Starting with a detailed kinetic model for methyl butanoate ($\text{CH}_3\text{CH}_2\text{CH}_2\text{C}(=\text{O})\text{OCH}_3$) oxidation, revisions are introduced to the C_0 - C_4 chemistry based on the recent Aramco mechanism 3.0. The resulting mechanism is combined with a short model for methyl crotonate, derived from a suitable reference mechanism. Several new classes of reactions are included and the rate constants of the existing reactions are revised based on various theoretical studies and analogies to reactions of similar species. Furthermore, the low-temperature chemistry of methyl crotonate has been implemented in the current study to extend the validity of the mechanism to lower temperatures. The resulting methyl crotonate combustion mechanism has been comprehensively validated using various experiments in the literature. In addition, experiments are performed using a heat flux burner at atmospheric conditions to measure the laminar burning velocities of methyl crotonate at different unburnt mixture temperatures (318, 338, and 358 K). The mechanism is found to reproduce the experimental data for high-temperature combustion of methyl crotonate satisfactorily. The mechanism is also found to predict the low-temperature ignition delays accurately. Sensitivity and path flux analysis are performed to delineate the importance of the different reaction classes in methyl crotonate chemistry. The current study presents a comprehensive mechanism for methyl crotonate combustion, along with a new set of experimental results complementing the existing experimental database in the literature.

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1. Introduction

Petroleum-based fuels constitute a major fraction of the world's total energy consumption [1]. The increased depletion rate of petroleum resources has resulted in a quest for alternative energy forms that are both eco-friendly and renewable. One such candidate is biodiesel, derived from the trans-esterification of vegetable oils and animal fat, and is considered as a prospective candidate to completely or partially replace diesel [2]. Biodiesel typically consists of long-chain methyl and ethyl esters (C_{14} - C_{22} esters), which are either saturated or unsaturated [3]. Studies of several large representative esters [4–11] such as methyl decanoate, methyl stearate, and methyl oleate demonstrated the ability of these fuels in predicting various aspects of biodiesel combustion such as the presence of a negative temperature coefficient (NTC),

an early CO_2 emission, and similarity in reactivity in comparison to biodiesel. Nevertheless, the oxidation kinetics of such large esters are highly intricate and often require prodigious computational and human effort in their development. Besides, the unavailability of rate coefficients for their oxidation pathways exacerbates the challenge of developing an accurate chemical kinetic mechanism.

In order to circumvent this complexity, the oxidation chemistry of smaller representative fuels having specific similarities to their larger counterparts can be investigated. Subsequently, reliable reaction mechanisms for larger ester fuels can be obtained based on the rate-rules developed for the smaller representative fuels. Thus, the focus has been shifted to the studies of smaller esters such as methyl butanoate (MB), a saturated C_5 methyl ester. Several experimental and modeling studies were reported considering MB as a representative of saturated methyl esters [12–16]. However, most of the varieties of biodiesel fuels contain a significant portion of unsaturated methyl esters rendering it important to understand the kinetics of the same [3]. The present work focuses on the combustion kinetics of methyl crotonate (Methyl but-2-enoate,

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MB2D), a C₅ methyl ester possessing an unsaturation on the second carbon.

The choice of MB2D is motivated by the fact that it is a simple unsaturated ester having structural similarity to MB and yet brings in the unsaturation present in real biodiesels. The C=C bond in MB2D exists near the ester group while the unsaturation occurs further away in real biodiesel. Nevertheless, it is important to explore the oxidation kinetics of small unsaturated esters in detail which would provide a platform to develop the kinetics of large unsaturated esters. Besides, the availability of a wide range of experimental data in the literature [17–27] enables comprehensive validation of the MB2D combustion kinetics. These experiments include homogeneous reactors, premixed and non-premixed flames, and experiments measuring soot and NO_x, and is discussed below in the aforementioned order. Sarathy et al. [17] studied the oxidation of MB and MB2D under atmospheric conditions in a jet-stirred reactor (JSR). Results demonstrated similar reactivity for MB and MB2D, with MB2D exhibiting a greater tendency to produce soot precursors compared to MB. Gaïl et al. [18] performed further speciation measurements for the oxidation of MB and MB2D in a jet-stirred reactor at atmospheric pressures, over the temperature range 850–1400 K. In addition to the experiments, the authors added new reactions to their previous model to better represent MB2D oxidation. Bennadji et al. [19,20] determined ignition delay times (τ_{ig}) for MB2D and several other short esters using shock tubes at pressures of 7–9.65 atm, temperatures of 1280 to 1930 K, and equivalence ratios of 0.25, 1.0, and 2.0. The study employed computer-generated kinetic models to compare the simulated τ_{ig} to the experiments.

In addition to homogeneous reactor experiments, Sarathy et al. [17] investigated the oxidation of MB and MB2D under atmospheric conditions using an opposed flow diffusion flame burner. Further, Wang et al. [21] determined the laminar burning velocities (S_L) and extinction strain rates ($a_{2,E}$) for MB, MB2D, and methyl decanoate flames in counterflow configurations, at atmospheric pressure and elevated unburned mixture temperatures. These data were compared against those derived for the *n*-alkane flames of similar carbon number, to assess the effects of saturation, the length of carbon-chain, and the presence of an ester group. Yang et al. [22] studied the oxidation of MB2D, methyl methacrylate, and ethyl propenoate at 2.67 and 4 kPa under fuel-rich premixed conditions in a stainless steel McKenna burner. Niemann et al. [23] measured the auto-ignition temperature ($T_{2,i}$) and extinction strain rates ($a_{2,E}$) of MB, MB2D, ethyl propionate, biodiesel, and diesel using counterflow diffusion flame burner.

Feng et al. [24] investigated the NO_x production in MB and MB2D flames under both premixed and non-premixed conditions in a stagnation-flow configuration. A higher production of NO_x was observed in MB2D flames compared to MB, mainly due to the higher flame temperatures in the former. Sooting propensities of MB2D has also been examined under non-premixed conditions [25,26]. It was found that the ester produced lesser soot precursors when compared to its corresponding *n*-alkane. Also, peak concentrations of sooting precursors like C₂H₂, C₂H₄, and C₃H₃ were found to be at high levels in MB2D oxidation when compared to MB. Zhai et al. [27] recently investigated the pyrolysis of MB and MB2D and found higher amounts of unsaturated hydrocarbons being produced in the pyrolysis of the latter.

Complementing the experimental and modeling studies in the literature, several theoretical studies [28–31] explored the oxidation kinetics of MB2D to evaluate accurate bond energies and rate constants of reaction pathways. Despite the availability of a wide range of experimental and theoretical studies for the combustion of MB2D, not all these studies have been utilized for modeling the oxidation of MB2D. The present work aims to develop a novel comprehensive mechanism for the combustion of MB2D, utiliz-

ing the experimental and theoretical rate constant determinations available in the literature. Further, the validation database has also been expanded in the present study by obtaining new measurements with MB2D as fuel, contributing to both the experimental and modeling fronts.

The highlights of the present work are thus two-fold: (1) New experimental data for laminar burning velocities of MB2D measured in a heat flux burner at different unburned temperatures (318, 338, and 358 K) are presented and used to validate the newly developed mechanism. (2) Several new reaction pathways for the high-temperature oxidation of MB2D are introduced which were not available in existing models. One of our recent studies [32] investigated the ignition delays of MB2D/O₂/diluent mixtures in a rapid compression machine (RCM) at pressures of 20 and 40 bar, unburnt temperatures ranging 900–1074 K, and different equivalence ratios ($\phi = 0.25, 0.5$ and 1.0). The importance of low-temperature pathways was found to be notable in predicting the experimental trends [32]. Thus, the low-temperature chemistry of MB2D is incorporated in the present model making this study the foremost in exploring the oxidation of MB2D in a comprehensive manner.

Recently, Lele et al. [16] developed a comprehensive model for the oxidation of MB and validated the same for numerous experiments. Subsequently, a surrogate formulation was developed using MB and *n*-dodecane (*n*-C₁₂H₂₆) [33] to predict the combustion of actual biodiesel. To use MB2D as a representative of large unsaturated esters, it is inevitable to ascertain differences in the combustion behavior of MB2D to that of MB. The current study will be followed by a subsequent investigation analyzing the differences between combustion of MB and MB2D in-depth, covering ignition delays, premixed, and non-premixed flames.

This article is organized as follows. Section 2 discusses the experimental approach in obtaining the laminar burning velocities in a heat flux burner. A description of the different classes of reactions and their kinetics will be presented in Section 3. The experimental results of the current work and the results of model validation and their discussion will be presented in Section 4, following which the conclusions are summarized in Section 5.

2. Experimental methodology

Planar, adiabatic, premixed laminar flames were stabilized at atmospheric pressure on a perforated plate burner using heat flux method [34]. This flat-flame-based method has been extensively used by many research groups to accurately measure the laminar burning velocities (S_L) of gaseous and liquid fuels since it creates a stable and steady flame that, at suitable conditions, can be approximated as a one-dimensional, adiabatic and stretchless flame. A review of the method and its development, detailed descriptions of the experimental setup and data processing algorithms have been presented elsewhere [35], and only the relevant elements are outlined here.

The experimental apparatus [36] consists of a heat flux burner composed of a 2 mm-thick brass plate, which is integrated with the burner head, and perforated with 0.4 mm-diameter holes at a pitch of 0.47 mm. The burner plate has a nominal diameter of 3 cm and an effective perforation area of 6.9004 cm². It is surrounded by a heating jacket supplied with thermostatic water to keep its edge at a constant temperature of 368 K, while the temperature of the fresh gas mixture (T_g) is controlled by a separate heating system in the plenum chamber, supplied with water. In the present experiments, T_g has been fixed at 318, 338, and 358 K. These temperatures have been chosen in order to cover a wider range of equivalence ratios, which is constrained by the vaporization limit of the mixture. A third circulating water jacket ensures thermal insulation between the hotter top part of the burner head

and the bottom part/plenum chamber. The temperature of this additional water jacket has been kept the same as the plenum chamber. S_L was determined by varying the flow rate of the gas mixture until a uniform radial temperature distribution was achieved over the burner plate, recorded by fifteen E-type thermocouples (0.076 mm bare wire diameter) soldered in the burner plate. For this condition, the net heat flux from/to the burner plate is zero and the velocity of the unburned gas is equal to the adiabatic laminar burning velocity. S_L is a linear function of flow velocity provided that the flame is stable at adiabatic conditions. In this regard, Hermanns [37] found that the temperature difference between the burner plate and unburned gas mixture should be higher than 30 K (this value depends on the fuel mixture) to avoid flame instabilities. However, in the present experiments, this ΔT has been set to 10–50 K. Consequently, at $T_g = 338$ and 358 K the flames exhibited corrugation around adiabatic conditions. In these cases, S_L was determined by extrapolation from flat sub-adiabatic conditions. An example of evaluating S_L from sub-adiabatic flames and the associated uncertainties can be found in [35].

A mixing panel provides a controlled flow of the vaporized fuel and air to set the required equivalence ratio. The liquid fuel feeding system consists of a stainless steel cylinder, pressurized with nitrogen to feed the fuel through a Coriolis liquid mass flow controller into a Controlled Evaporator Mixer (CEM), both provided by Bronkhorst High-Tech. The CEM consists of a control valve, a mixing device, and a heat exchanger, the temperature of which is controlled by a temperature controller that is part of the system. The required liquid flow rate is metered to the set-point value by a control valve that forms an integral part of the CEM system. The total uncertainty in the liquid flow rate is a sum of 0.5 g/h and the stated flow repeatability of the liquid flow controller, which corresponds to 0.2% of the reading. Methyl crotonate (98% purity) was used as delivered from Sigma Aldrich in sealed bottles. Synthetic air was delivered by AGA (21% O₂, 1% relative uncertainty). The required air flow rates were metered by means of a Bronkhorst High-Tech digital thermal mass flow controller. Air was used as the carrier gas to stimulate the evaporation process as a mixing component and to transport the vapor. The gas flow meter was calibrated before measurements using a MesaLabs Definer 220 positive displacement calibrator. The total uncertainty of the air flow rate is the sum of 1% stated accuracy of the calibrator and the stated flow repeatability of the mass flow controller, which corresponds to 0.2% of the set operating conditions. In order to prevent fuel condensation throughout the fueling system, an electrically heated tube was used to connect the evaporator to the plenum chamber. The temperature of the heated pipe was set equal to T_g .

The experimental laminar burning velocities measured in the present study and the associated uncertainties are listed out in supplementary material (see Table S4 in supplementary document). The uncertainty associated with the unburnt gas temperature (T_g) is ± 1 K. A detailed analysis and quantification of experimental uncertainties were reported earlier [35] and the overall accuracy of S_L in the present measurements was estimated to be better than ± 0.6 cm/s. The equivalence ratio of the mixtures was varied from 0.7 to 1.3 with a maximum uncertainty of 0.017.

3. Kinetic mechanism development

3.1. Reference mechanism and MB2D chemistry

A comprehensively validated kinetic mechanism for the oxidation of MB by Lele et al. [16] (Lele model) is chosen as the reference mechanism. MB2D is an intermediate that appears in the MB chemistry (denoted as MB2D) and its most important oxidation pathways such as unimolecular decomposition and radical addition (O and HO₂ addition) are present in the Lele model. How-

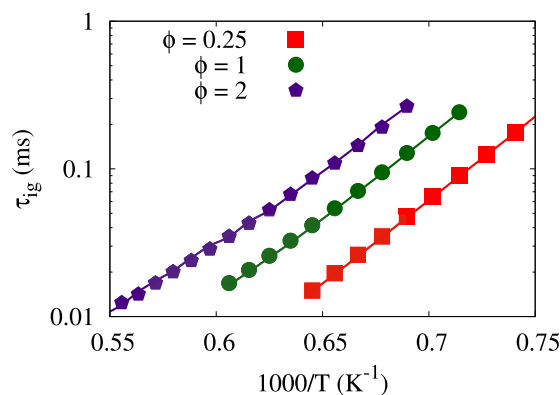
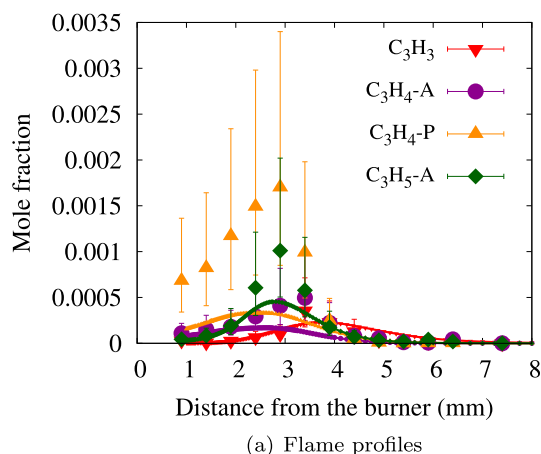


Fig. 1. Comparison of computed τ_{ig} between the Gail model [18] (Symbols) and compact MB2D sub-model (Solid lines) for MB2D/air mixtures at 8 atm.

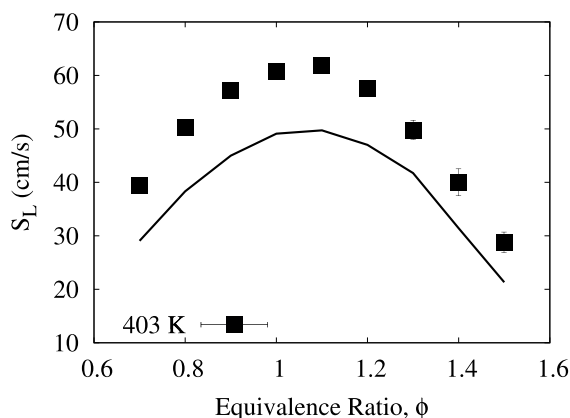
ever, to predict all the combustion characteristics of MB2D, its kinetics needs to be developed exhaustively. To begin with, an MB2D mechanism consisting of 301 species and 1516 reactions [18] (Gail model) has been chosen as the fuel-specific reference mechanism from the literature. The Gail model was originally validated for JSR and counterflow diffusion flame configurations [18]. Being the earliest MB2D model, the Gail model was used to simulate different studies which includes high- and intermediate-temperature ignition delay studies [20,32] and laminar premixed and diffusion flames [21]. The ability of the Gail model to reasonably predict these configurations coupled with the absence of a comprehensively validated MB2D mechanism, fueled the choice of the Gail model as the MB2D reference mechanism.

A compact mechanism for MB2D oxidation (90 species and 522 reactions) is derived by reducing the Gail model [18] using DRGEP technique [38]. The reduction is carried out using Chemkin Pro [39] and the configuration for reduction is a homogeneous isochoric reactor for temperatures of 1400 and 1600 K, pressures of 10 and 40 bar, and ϕ of 0.5, 1.0, and 2.0. The reduction targets include ignition delays (τ_{ig}), final gas temperature, pressure, mole fractions of CO₂, H₂O, and O₂ and maximum mole fractions of CO, OH, and CH radicals with an allowed relative error of 5%. Figure 1 shows the comparison of τ_{ig} for MB2D between the Gail model and the compact MB2D mechanism at 8 atm and $\phi = 0.25$, 1.0, and 2.0. A very good agreement is observed between the Gail model and the compact MB2D sub-model. This MB2D sub-model is then augmented to the Lele model [16] in a consistent manner to obtain a combined MB-MB2D mechanism.

Regardless, the MB-MB2D mechanism underpredicts the species profiles (see Fig. 2(a)) in a premixed MB2D flame [22] and laminar burning velocities [21] (see Fig. 2(b)) when compared against the experimental results in literature. The model underpredicts important C₃ intermediates like allene (C₃H₄-A), propyne (C₃H₄-P), and allyl radical (C₃H₅-A) (see Fig. 2(a)). Moreover, at $\phi = 1.1$, the MB-MB2D mechanism underpredicts the laminar burning velocity by 12.2 cm/s (Fig. 2(b)). Figure 3 shows the sensitivity analysis for laminar burning velocity of rich MB2D/air mixtures ($\phi = 1.1$), using the MB-MB2D mechanism. A positive sensitivity coefficient for a reaction indicates an increase in the burning velocity when the reaction rate increases. The sensitivity chart clearly reveal the importance of the base chemistry and the high-temperature chemistry of MB2D (H-abstractions by H and OH in MB2D) in predicting laminar burning velocities. This sensitivity analysis suggests a revisit to the base chemistry and the fuel-specific chemistry of MB2D, to improve the accuracy of the mechanism. Gail et al. [18] in their study developed the Gail model considering the high-temperature oxidation pathways of MB2D. Besides, the Gail model is reduced in the present study at high-temperature regimes (1400 and 1600 K,



(a) Flame profiles



(b) laminar burning velocities

Fig. 2. Results of (a) species profiles in premixed MB2D flame and (b) S_L for MB2D/air mixtures computed using the MB-MB2D mechanism before any revisions. Symbols correspond to experimental data [21,22] in the literature and lines (and lines with symbols) represent simulated results.

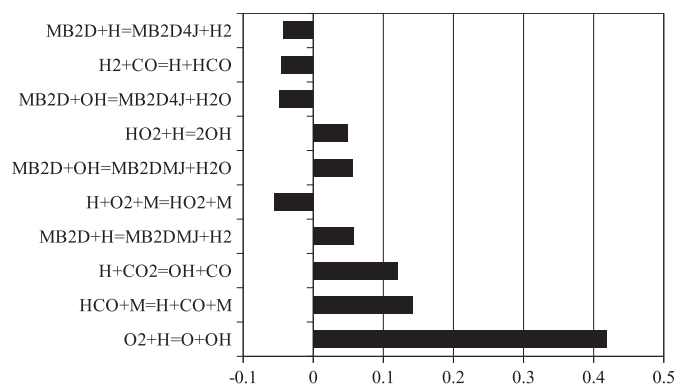


Fig. 3. Sensitivity analysis for the laminar burning velocity of MB2D/air mixtures at $\phi = 1.1$ using MB-MB2D mechanism before revisions.

10 and 40 bar, homogenous isochoric reactor). Therefore, it is essential to incorporate the low-temperature oxidation pathways of MB2D to make the present study a comprehensive one.

The base chemistry (C_0 – C_4) in the MB-MB2D mechanism is taken from the Lele model [16], which was originally derived from a well studied *n*-dodecane mechanism [40]. Owing to the availability of newer studies on short alkanes [41–49] (Aramco Mechanism 3.0), the base chemistry (C_0 – C_4) of the MB-MB2D mechanism has been updated using the Aramco mechanism [49] consistently. This allows to accommodate the improved short-alkane kinetics and

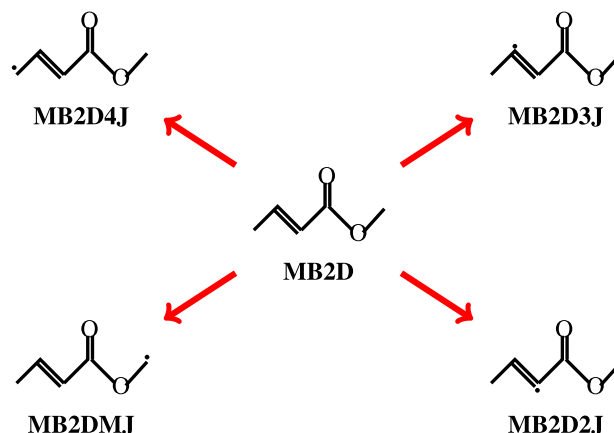


Fig. 4. Different products of four H-abstraction reactions in MB2D.

thereby improve the ability of the mechanism in predicting intermediate species concentrations. Revisions implemented in the base chemistry are limited to the reactions involving species present in both MB-MB2D mechanism and Aramco Mechanism.

3.2. Revisions in the fuel-specific chemistry of MB2D

Gaïl et al. [18] considered two sites in MB2D for H-abstraction producing MB2D4J and MB2DMJ (see Fig. 4), and assigned equal rate constants for the abstraction at both the sites. However, MB2D4J is more stable in comparison to MB2DMJ, due to the existence of resonance delocalization of electrons. Moreover, the H-abstraction pathways to produce the vinylic radicals (MB2D2J and MB2D3J in Fig. 4) are not considered in the Gaïl model. The vinylic radicals, though produced in minor amounts, can be important in predicting various C_3 species. To account for the above observations and to ensure kinetic consistency, it is imperative to revisit the oxidation kinetics of MB2D in the MB-MB2D mechanism to better the calibre of the model.

3.2.1. Unimolecular decomposition pathways

The unimolecular decomposition pathways are cardinal in predicting the high-temperature combustion behavior of any fuel [50]. Zhai et al. [27] studied the pyrolysis of MB2D in a flow reactor and adopted analogies for different decomposition reactions (C–C and C–O bond cleavage). In this study, the rate expressions for the decomposition reactions involving the cleavage of all C–C bonds and (CO)–OCH₃ bond are adopted from [27].

Zhai et al. [27] adopted the analogy of dimethyl carbonate decomposition ($CH_3OC(=O)O-CH_3$, bond dissociation energy of 89.6 kcal/mol [51] for O–CH₃ bond) for the cleavage of O–CH₃ bond in MB2D. However, the O–CH₃ bond in MB2D is found to have a lower bond dissociation energy (85.8 kcal/mol) [28], which is similar to that of the corresponding bond dissociation energy in MB (86 kcal/mol) [28]. Moreover, it is observed that the drop in magnitude of the rate constant for dimethyl carbonate decomposition is delayed toward lower temperatures, when compared to the corresponding rate constant for MB2D decomposition in Gaïl model [18] and MB decomposition in the recent Lele model [16]. As a result, this decomposition pathway in Zhai et al. [27] is observed to contribute a significant fraction (15%) in the consumption of MB2D at intermediate temperatures (1200 K and 40 bar). The similar pathway in the MB2D chemistry in Gaïl model and the MB chemistry in Lele model is found to contribute only about 7% and 1% in the consumption of MB2D and MB respectively, at similar conditions.

Therefore, owing to the lack of additional information on accurate rate constant determinations for this important reaction, the high temperature rate constant has been adopted from Gail et al. [18]. Further, the low temperature rate constant for this reaction has been reduced by a factor of 5 so as to retain the similarity to MB decomposition at lower temperatures. Additional investigation of the unimolecular decomposition pathways of MB2D is required to establish accurate temperature and pressure dependence of these pathways.

3.2.2. H-abstraction by H, OH, and other radicals

The current study considers the H-abstraction from MB2D at all four possible locations. The different products of H-abstraction pathways are shown in Fig. 4. Li et al. [29] theoretically investigated the kinetics of H-abstraction from MB2D by H using MS-CVT/SCT theory. Further, Zhou et al. [30] explored the kinetics of H-abstraction from MB2D by OH based on CBS-QB3 and QCISD(T)/CBS methods. The rates for these reactions are taken from the aforementioned studies in the MB-MB2D mechanism. The rate constants for the H-abstraction by other species such as O, O₂, CH₃, and CH₃O, are obtained using analogies to MB, methyl propenoate, [16] and alkene (propene, 2-butene, and isobutene) kinetics [49,52,53].

At intermediate-to-low temperatures (1100–800 K), the concentration of meta-stable radicals such as HO₂ and CH₃O₂ dominates when compared to reactive radicals like H atoms. Hence, H-abstractions by HO₂ and CH₃O₂ turns out to be significant and influence the reactivity of MB2D at these temperature ranges. However, no theoretical or experimental studies are available for such reactions. Therefore, the current study obtains these rate constants from the corresponding reactions of analogous molecules. The rate expressions for H-abstraction (by HO₂ and CH₃O₂) producing MB2D4J are obtained based on analogies to similar reactions of isobutene [49,54], and those producing MB2DMJ are based on analogies to MB chemistry [16]. The H-abstractions at vinylic sites are not considered, as these reactions with higher activation barriers are not significant compared to other channels at low-to-intermediate temperatures.

3.2.3. Isomerization reactions

The reactions involving isomerization of different fuel radicals have not been considered in the Gail model [18]. In this study, the isomerization of MB2D4J to MB2DMJ is included, and the rate constant for this reaction is chosen based on an analogy to the 1–6 H-shift in hex-2-en-6-yl radical [55]. The isomerization of MB2D4J to MB2D2J and MB2D3J are added based on analogies to similar reactions of allyl radical (C₃H₅-A) [49]. The three isomerization reactions involving MB2DMJ, MB2D2J, and MB2D3J are prescribed using analogies to similar reactions in methyl propenoate chemistry [16].

3.2.4. β -scission of fuel radicals

The fuel radicals (see Fig. 4) produced from the H-abstractions readily undergo β -scission at higher temperatures (> 1200 K). The β -scission of MB2DMJ, MB2D2J, and MB2D3J produce various C₃ and C₄ species, and the rate expressions for these decomposition pathways are derived using analogous reactions of MB and methyl propenoate (CH₂=CHC(=O)OCH₃) [14]. The presence of a C=C bond at the β -position in MB2D4J impedes the β -scission. Nevertheless, the scission of C–H bond at the β -position is considered in analogy to similar reaction in propene chemistry [49]. Also, the addition of O atom to MB2D4J followed by a β -scission is considered using analogies to allyl pathways [40].

The β -scission of the C–H bond in MB2D4J gives rise to MB2,3D (CH₂=C=CHCOOCH₃), which resembles the structure of allene (C₃H₄-A). On account of this similarity, the consumption path-

ways for MB2,3D in the MB-MB2D mechanism include unimolecular decomposition, H-abstraction, and radical addition pathways. These pathways are assigned rate constants based on the analogous reactions involving similar species like allene [40,49] and MB2D [27,29,30].

3.2.5. Radical addition reactions

The presence of a C=C bond in MB2D can open up further avenues for oxidation of MB2D by inviting radical additions across the double bond, followed by β -scission. The current study accounts for the addition of H, OH, O, HO₂, and CH₃ on MB2D. The pressure dependent rate expressions for different OH addition channels are adopted from the theoretical study by Zhou et al. [30]. The additions of H atoms on MB2D are included based on kinetics developed as part of 2-pentanone chemistry [56]. The pathways involving addition of CH₃ are incorporated based on analogies to similar reactions of CH₂=CHCHO [49], along with the reactions involving CH₃ addition on MB2D from Zhai et al. [27]. The HO₂ addition reactions, which are already present in the MB-MB2D mechanism as a part of the low-temperature oxidation chemistry of MB, comes from [16].

The reactions involving the addition of O atom across the double bond producing various radicals are found to influence the prediction of τ_{ig} and S_L at fuel-lean conditions (see Fig. S1 and S2 in supplementary material). The rate constants for these reactions are adopted from similar pathways in 1,3-butadiene (C₄H₆) oxidation [49]. Nevertheless, the higher stability of MB2D+O adduct compared to C₄H₆+O adduct, emanating from the presence of a more stable secondary radical (CH₃CHCH(Ö)COOCH₃ and CH₃CH(Ö)CHCOOCH₃) and the presence of an electronegative C=O group in the former has to be accounted for. In order to accommodate these effects, the rate expressions of the O addition pathways are increased by a factor of 10 from those of C₄H₆+O system. Further studies are required to obtain accurate rate expressions for this reaction class.

3.2.6. The R+O₂ system and low-temperature chemistry

The reactions involving successive addition of O₂ to the fuel radical (R) are crucial in predicting the reactivity of MB2D at intermediate-to-low temperatures (1100 to 800 K). The prominence of these reactions are found elsewhere in the literature [32]. Among these is a chain branching sequence that proceeds to produce two reactive OH radicals, thereby increasing the reactivity of MB2D. Despite their importance, these reaction classes were found to be absent in the Gail model [18], and thus in the MB-MB2D mechanism. At this point, the MB-MB2D mechanism has been employed to simulate the intermediate temperature ignition delays of MB2D/O₂/diluent mixtures from literature [32], and the results are portrayed in Fig. 5. The predictions clearly reflect the low reactivity of the MB2D chemistry in MB-MB2D mechanism towards lower temperatures. Therefore, it is necessary to include the R+O₂ system of reactions to the MB-MB2D mechanism to boost the reactivity of the MB2D kinetics at intermediate and lower temperatures.

The low-temperature branching sequence commence with the addition of O₂ on MB2D radical (R+O₂ → ROO) to produce a fuel-peroxy radical [50]. The O₂ addition reactions considered in the current study are shown in Fig. 6. Corresponding reactions for MB2D2J and MB2D3J are not considered in this study, as these radicals are produced in insignificant quantities at lower temperatures. Subsequently, the ROO can undergo (i) a thermal decomposition back to R+O₂ at higher temperatures, and the fuel radical can undergo a β -scission or other high-temperature reactions as described above (ii) a concerted elimination reaction, producing a stable olefinic species and a meta-stable HO₂ radical (iii) an internal H-migration proceeding through a cyclic intermediate to produce a hydroperoxy radical (ROO → QOOH). Further, QOOH can ei-

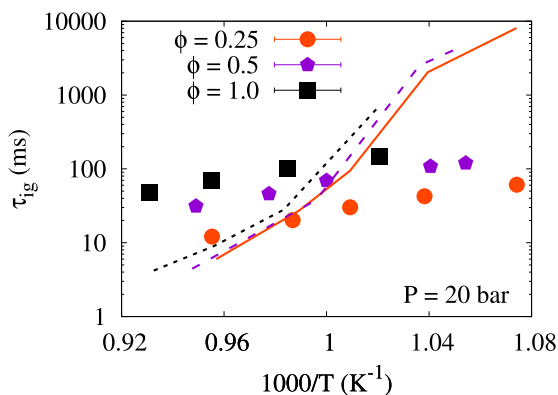


Fig. 5. Simulated results (lines) of τ_{ig} using the MB-MB2D mechanism (without low-temperature chemistry) compared against the experimental data [32] (symbols) for MB2D/O₂/diluent mixtures at 20 atm.

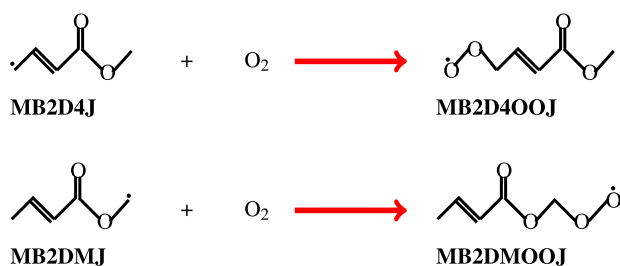


Fig. 6. Addition of O₂ on different MB2D radicals considered in the present work.

ther undergo (i) a chain-propagating epoxidation reaction producing one OH radical or (ii) a second O₂ addition to QOOH, followed by consecutive decompositions to produce two OH radicals, thus enhancing the reactivity of MB2D via a chain-branching sequence.

The low-temperature kinetics of MB2D was investigated and validated in our recent work [32]. The scheme of reactions were found to accurately capture the reactivity of MB2D. Hence, the low-temperature reaction classes of MB2D are incorporated from our earlier work [32] in the MB-MB2D mechanism.

The nomenclature of all fuel-specific species have been elucidated in the supplementary material (see Table S1). The proposed MB-MB2D mechanism, thermodynamic data, and transport properties have also been included as part of supplementary information.

4. Results and discussion

This section presents the validation results of the MB-MB2D mechanism for a wide range of experiments. The discussion includes ignition delay times at high and intermediate temperatures, laminar burning velocities, flame profiles under premixed and non-premixed conditions, followed by auto-ignition and extinction studies. In addition, the laminar burning velocities of MB2D/air mixtures measured in this work are presented and used for assessment of the kinetic model. The high-temperature ignition delay times, laminar burning velocities, and auto-ignition temperatures were computed using FlameMaster [57] (version 4.1). The simulations for all other configurations were performed using Chemkin Pro [39] (version 19.1). Table 1 lists the experiments used to validate the MB-MB2D mechanism along with the configurations used to simulate the measured data.

4.1. Ignition delay times

Ignition delay times (τ_{ig}) of MB2D mixtures, measured in a shock tube or a rapid compression machine (RCM) can be used to

understand the accuracy of the reaction kinetics of MB2D in the MB-MB2D mechanism in the absence of transport. The shock tube and RCM experiments (experimental heat losses are accounted for) are modeled using a homogeneous, isochoric, 0-D reactor model.

4.1.1. High-temperature ignition delays

The high-temperature ignition delays of MB2D/O₂/Ar mixtures were measured by Bennadji et al. [19] using a shock tube. The experiments were conducted at compressed pressures of 7.5–9 atm, temperature ranging from 1280 to 1930 K, and at different equivalence ratios ($\phi = 0.25, 1.0$, and 2.0). The study observed an increase in τ_{ig} (decrease in reactivity) with an increase in ϕ , similar to alkane combustion at higher temperatures. Figure 7 reports the experimental τ_{ig} from Bennadji et al. [19], along with the modeling results (at $P = 8$ atm) obtained using the Gail model [18] and the present MB-MB2D mechanism.

The MB-MB2D mechanism successfully predicts the experimental trend in τ_{ig} for all conditions. The model (solid line) shows a significant improvement in predicting the τ_{ig} at fuel-lean conditions (Fig. 7(a)), over the Gail model [18] (dashed line). Beside definite improvements at fuel-lean condition, minor improvements are observed within the data scattering in the predictions of τ_{ig} for stoichiometric mixtures (Fig. 7(c)) by the MB-MB2D mechanism. Thus, the MB-MB2D mechanism is found to exhibit major improvements thereby accurately reproducing the high-temperature ignition delays, in comparison to the existing model [18].

A sensitivity analysis for the ignition delay time is performed using the MB-MB2D mechanism at $\phi = 0.25$, as shown in Fig. 8. The analysis has been carried out by multiplying the rate constant of each reaction by a factor of 2, and observing the change in ignition delay time, before and after multiplication. A positive sensitivity coefficient denotes an increase in τ_{ig} , as the rate of reaction increases. The τ_{ig} is observed to be most sensitive to the unimolecular decomposition of MB2D, and increasing the rate of this reaction decreases the ignition delays. This is ascribed to the direct production of CH₃ radicals from the stable fuel (MB2D), thereby increasing the reactivity. In addition, the ignition delays are found to be sensitive to H-abstractions by H, O, and OH. These reactions reveal positive sensitive coefficients, as, they proceed consuming the reactive radicals like H and OH.

In order to understand the consumption of MB2D in a homogeneous isochoric reactor, a path flux analysis is performed for the MB2D/O₂/Ar mixtures at $\phi = 0.25$, $T = 1400$ K, and $P = 8$ atm, using the MB-MB2D mechanism (See Fig. 9). The pathways highlighted in blue are the new reactions considered in this study and the pathways shown in black arrows are originally from the Gail model and are revised in the current study. It is observed that, about 48% of MB2D is consumed through H-abstraction reactions producing four MB2D radicals (see Fig. 4 for different MB2D radicals) and 19% through the unimolecular decomposition of MB2D. The MB2D2J and MB2D3J radicals are produced in minor quantities (4% and 2% respectively), as the vinylic sites in MB2D offers a higher activation barrier for H-abstractions [29,30] compared to the other abstraction sites. The presence of a C=C bond in MB2D invites a significant fraction (31%) of MB2D to be consumed through O-addition reactions. These reactions involve the formation of an adduct, followed by a decomposition producing smaller radicals.

Considering the subsequent consumption pathways of the products formed above, MB2DMJ radical simply undergoes a β -scission (100%) whereas, a significant fraction (88%) of MB2D4J isomerizes to MB2DMJ. The C=C bond at the β position discourages any carbon-carbon β -scission in MB2D4J. A small fraction (6%) of MB2D4J undergoes a second H-abstraction to produce MB2,3D, which, is primarily consumed via radical addition pathways (74%).

Table 1

List of experiments used to validate the MB-MB2D mechanism and the configurations used to simulate the same.

Experiments	Configuration for simulations
Shock tube ignition delays	Homogeneous, isochoric, and adiabatic reactor (FlameMaster [57])
RCM ignition delays	Homogeneous reactor with imposed volume profile (Chemkin [39])
laminar burning velocities	Unstretched, premixed, freely propagating flame (FlameMaster [57])
Burner stabilized flame profiles	Premixed, laminar, burner-stabilized flame (Chemkin [39])
Counter-flow diffusion flame profiles	Non-premixed, opposed-flow flame (Chemkin [39])
Auto-ignition temperatures	Non-premixed, opposed-flow flame (FlameMaster [57])
Extinction strain rates	Extinction of diffusion opposed-flow flame (Chemkin [39])

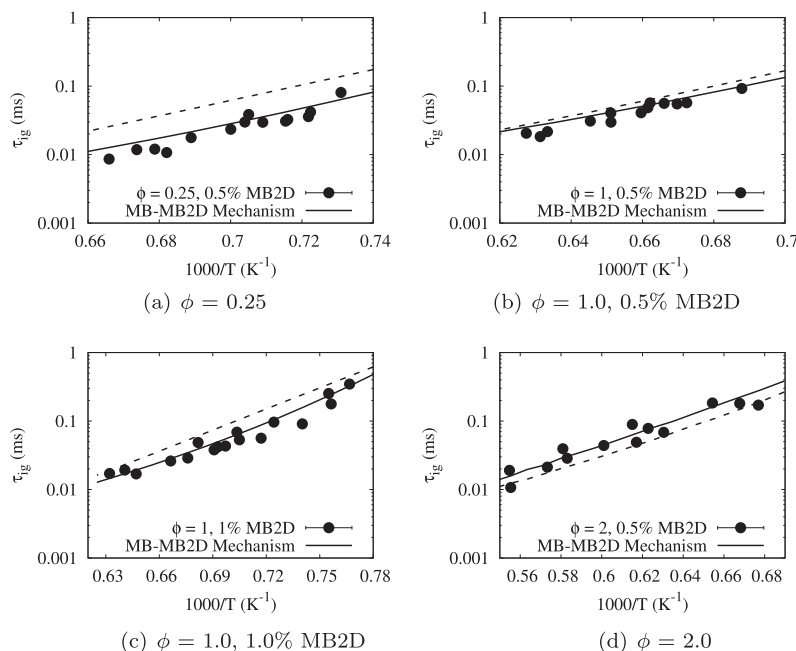


Fig. 7. Ignition delay times (τ_{ig}) of MB2D/O₂/Ar mixtures in a shock tube at $P = 8$ atm and (a) $\phi = 0.25$, (b) $\phi = 1.0$ and 0.5% MB2D, (c) $\phi = 1.0$ and 1.0% MB2D, and (d) $\phi = 2.0$. Symbols correspond to experimental data from Bennadji et al. [19] and the lines represent simulated results using the MB-MB2D mechanism (solid lines) and Gail model [18] (dashed lines).

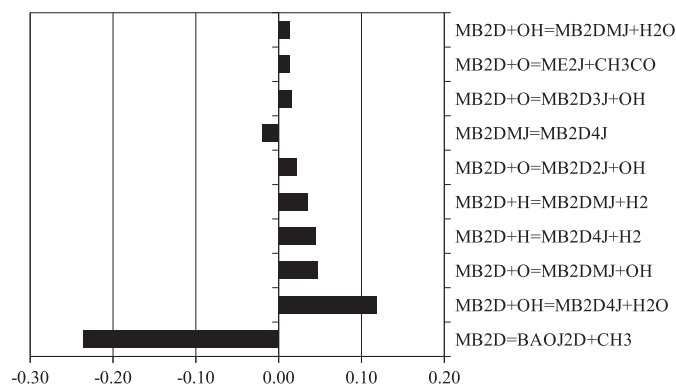


Fig. 8. Sensitivity analysis for the ignition delay times of MB2D/O₂/Ar mixtures at $\phi = 0.25$, $T = 1400$ K, and $P = 8$ atm, using the MB-MB2D mechanism. The sensitivity coefficients of the reactions involving fuel-specific species alone are shown. The analysis has been carried out using a sensitivity factor of 2. A positive sensitivity coefficient denotes an increase in τ_{ig} , as the rate of reaction increases.

To understand the effect of revisions in increasing the reactivity of MB2D at $\phi = 0.25$, a path flux analysis is performed using the version of MB-MB2D mechanism before revisions for the same conditions as reported in Fig. 9, and is depicted in Fig. 10. The effect of revisions introduced in the MB-MB2D mechanism on the reaction fluxes are shown in Table 2. It is to note that the revised O-addition pathways [49] in the MB-MB2D mechanism pro-

ceeds to produce numerous small radicals like H, HCCO, CH₃CO, and CH₃O, thereby contributing to the production of a reactive radical pool. Thus, the increase in the consumption of MB2D through O-addition channels (from 11% before revisions to 31% after revisions) increases the reactivity of the mechanism at $\phi = 0.25$, after the revisions are introduced.

4.1.2. Intermediate-temperature ignition delays

One of our recent works [32] performed an experimental and kinetic investigation on the intermediate-temperature ignition delays of MB2D/O₂/diluent mixtures in an RCM. The experiments were performed at pressures of 20 and 40 bar, temperatures ranging 900 to 1074 K, and three equivalence ratios ($\phi = 0.25, 0.5$, and 1.0).

In order to investigate the modeling capability of the MB-MB2D mechanism at lower temperatures, the mechanism has been employed to simulate the RCM experiments of MB2D [32]. The heat losses in the RCM are accounted by incorporating a non-reactive volume profile during the simulation, for each experimental condition. This procedure has been explained elsewhere in the literature [32,58,59]. Figure 11 presents the RCM ignition delays along with the simulated results at 20 and 40 bar.

The MB-MB2D mechanism is found to predict the trends in RCM experiments very well, and the predictions are found to be accurate overall. A maximum deviation of about 61% is observed at $P = 20$ bar, $T = 980$ K and $\phi = 1.0$. At 40 bar, a maximum deviation of about 33% is found to occur at $T = 905$ K and $\phi = 0.5$.

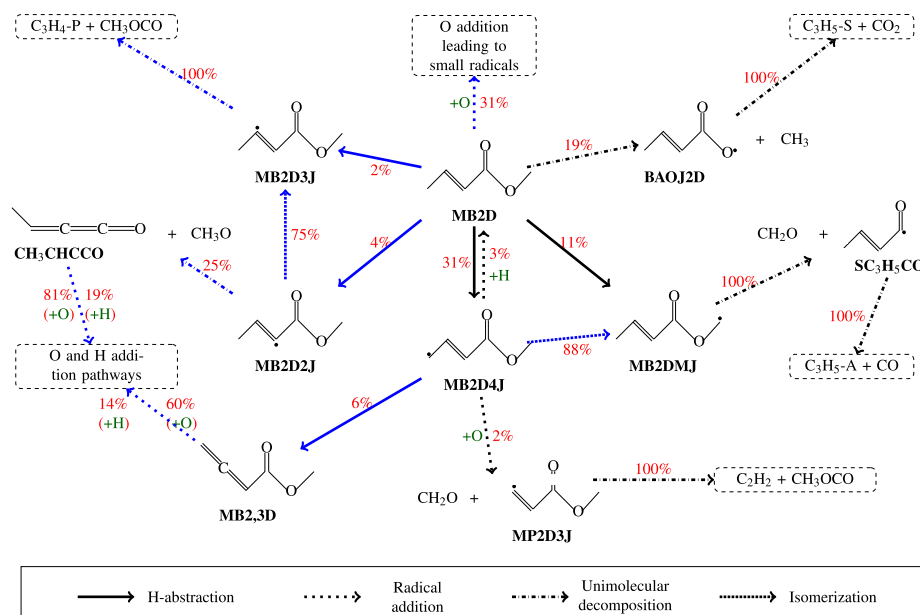


Fig. 9. Path flux analysis for the MB2D/O₂/Ar mixtures at $\phi = 0.25$, $T = 1400$ K and $P = 8$ atm in a isochoric, homogeneous reactor using the MB-MB2D mechanism. The pathways highlighted in blue are the new reactions considered in this study and the pathways in black arrows are originally present in the Gail model and are revisited in this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

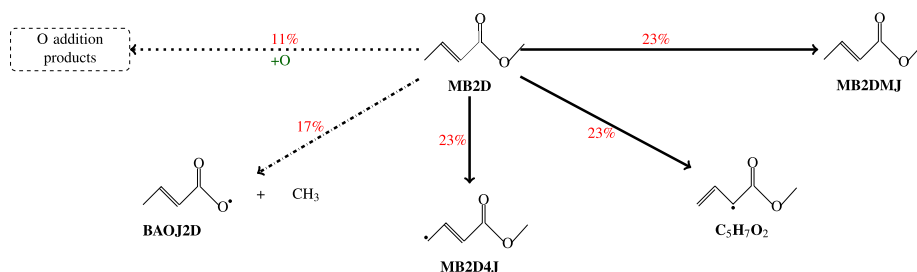


Fig. 10. Path flux analysis for the MB2D/O₂/Ar mixtures at $\phi = 0.25$, $T = 1400$ K and $P = 8$ atm in a isochoric homogeneous reactor using the MB-MB2D mechanism before any revisions. The definitions of different arrows are the same as that of Fig. 14.

Table 2

Changes in reaction fluxes after revisions and the corresponding effects on predictions for the MB2D/O₂/Ar mixtures at $\phi = 0.25$, $T = 1400$ K and $P = 8$ atm in a isochoric homogeneous reactor.

Reaction classes	Reaction flux (%)		Effect of revisions
	Before revisions	After revisions	
Unimolecular decomposition of O-CH ₃ bond [78]	19	19	Negligible change
O-addition pathways [49]	11	31	Increased production of small radicals like H, HCCO, and CH ₃ O (Reduces τ_{ig})
H-abstraction pathways [29,30]	46	48	The amounts of different fuel radicals change as opposed to equal production (23%) of fuel radicals
Isomerization (MB2D4J → MB2DMJ) [55]	-	88	Isomerization of MB2D4J dominates as opposed to O-addition in Gail model

No negative temperature coefficient (NTC) region is observed in the predictions of MB-MB2D mechanism at both 20 and 40 bar, upto 750 K (see Fig. S3), which is also supported by the experiments [32].

A sensitivity analysis for the ignition delay time is performed at $\phi = 0.5$, $P = 40$ bar, and $T = 900$ K, and is presented in Fig. 12. The analysis is performed in a similar manner, as done for high-temperature ignition delays (see Fig. 8). The ignition delay time is found to be sensitive toward the pathways involving low-temperature species such as cyclic ethers and QOOH (MB2DCY4OM and MB2DMOOH4J respectively in Fig. 12), along with other high-temperature pathways. It is noted that the H-abstraction from MB2DCY4OM by OH reveals a positive sensitivity coefficient, while that by HO₂ reveals a negative sensitivity coefficient. This is because the meta-stable HO₂ in the latter produces H₂O₂, which can

decompose producing two OH radicals, increasing the reactivity. However, the consumption of reactive OH to produce a less reactive fuel-specific radical (MB2D4JMDO) in the former, tends to decrease the reactivity of MB2D.

The ability of the newly developed mechanism in predicting the RCM ignition delays of MB2D [32], combined with its ability to predict the high-temperature ignition delays [19], substantiates the accuracy of the comprehensive MB-MB2D mechanism at low and high temperatures.

4.2. Flame measurements

The flame measurement studies include laminar burning velocities, premixed burner-stabilized flame profiles, and counter-flow diffusion flame profiles. The simulations for these flame studies are performed with the diffusion equation including thermal diffusion

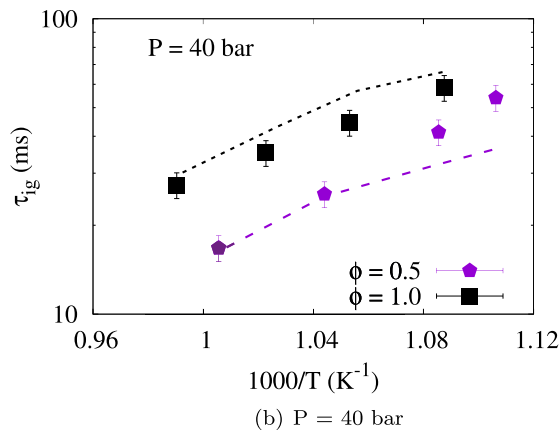
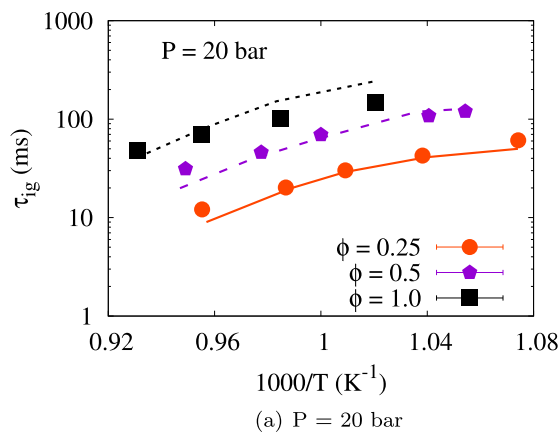


Fig. 11. Ignition delay times (τ_{ig}) of MB2D/O₂/diluent mixtures in an RCM at $\phi = 0.25, 0.5$, and 1.0 and (a) $P = 20$ bar, and (b) $P = 40$ bar. Symbols correspond to experimental data from Vallabhuni et al. [32] and the lines represents simulated results using the MB-MB2D mechanism.

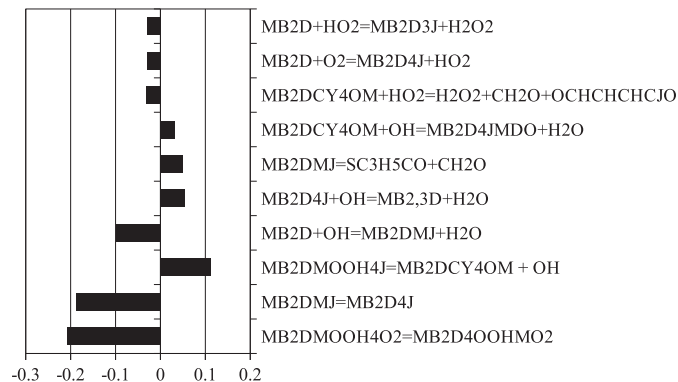


Fig. 12. Sensitivity analysis for the ignition delay times of MB2D/O₂/diluent mixtures at $\phi = 0.5$, $T = 900$ K, and $P = 40$ bar, using the MB-MB2D mechanism. The sensitivity coefficients of the reactions involving fuel-specific species alone are shown.

and multi-geometric transport model and the energy equation accounting for radiative heat loss.

4.2.1. Laminar burning velocities

Assessment of consistency in experimental data

The laminar burning velocities of MB2D/air mixtures at atmospheric pressures under different initial temperatures (318, 338, and 358 K) and equivalence ratios ($\phi = 0.7$ – 1.3) have been measured in a heat flux burner as part of this work (see Section 2). The measured laminar burning velocities along with the associated un-

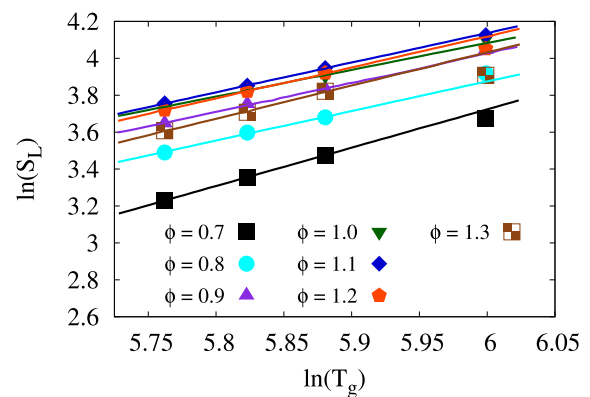


Fig. 13. Results of laminar burning velocities (S_L) of MB2D/air mixtures at different initial temperatures (T_g) and equivalence ratios (ϕ). Symbols: current work. Symbols corresponding to $\ln(T_g) = 6$: Wang et al. [21]. Lines: linear fit constructed based on from present work.

certainities can be found in the supplementary material (see Table S2). The laminar burning velocities of MB2D are depicted on a log-log plot as shown in Fig. 13. Owing to the availability of additional experimental data for laminar burning velocities of MB2D [21], it is worthy to discuss the consistency in the available experimental data before proceeding to the model validation.

The most utilized correlation describing the dependence of S_L on initial temperature (T_g) [60] is:

$$S_L = S_L^0 \left(\frac{T_g}{T_g^0} \right)^\alpha \quad (1)$$

where, the superscript '0' refers to a reference condition (in this case, 318 K). The temperature dependence, expressed by the exponent α , varies with ϕ . Such dependence can be expressed by plotting S_L as a function of T_g on a log-log scale, as reported in Fig. 13. In this way, the present experimental data can be compared with the experimental results measured by Wang et al. [21] at $T_g = 403$ K ($\ln(T_g) = 6$ in Fig. 13), using the counterflow twin flame method with nonlinear extrapolation to zero stretch. In Fig. 13, the lines represents a linear fit for the present experimental data at different ϕ .

When extrapolating the linear fit to higher temperatures, it is observed that the present data are consistent with the data from literature for $\phi \leq 1.1$, while at higher ϕ the agreement deteriorates (see Fig. 13). One reason for the observed inconsistency at fuel-rich conditions could be due to the experimental uncertainty in equivalence ratio, associated with the dataset available in the literature [21]. For instance, at $\phi = 1.3$, the uncertainty associated with the equivalence ratio is estimated to be 0.017 for the present experiments, while it is 0.075 (shown in Fig. 14(a) at $\phi = 1.3$ alone for better visibility) for the experimental data in literature [21].

Assessment of kinetic model

In order to establish the accuracy of the newly developed MB-MB2D mechanism, the experiments from present study and that of Wang et al. [21] are simulated using this mechanism. Figure 14(a) presents the experimental laminar burning velocities (symbols) along with simulated results (lines). The model is found to predict the laminar burning velocities of MB2D accurately at all conditions explored in this study. A maximum deviation of 2.84 cm/s is observed at $\phi = 0.7$ and $T_g = 358$ K. Further, the power exponent (α) is derived from the present experiments, and is shown as a function of ϕ in Fig. 14(b) (symbols). This provides a stringent test for the validation of a detailed kinetic mechanism. The MB-MB2D mechanism is found to follow the experimental trend in α within the experimental uncertainties (line in Fig. 14(b)). Overall,

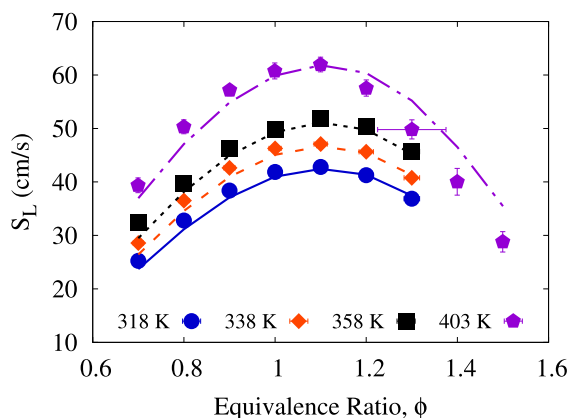
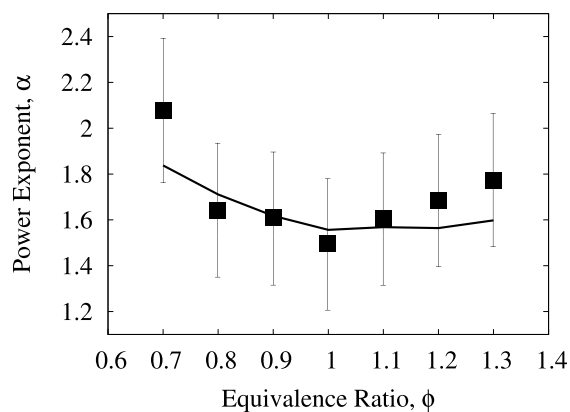
(a) laminar burning velocities (S_L)(b) Power exponent (α)

Fig. 14. Experimental and modeling results of a.) laminar burning velocities vs. ϕ at $T_g = 318, 338$, and 358 K from the present study and $T_g = 403$ K from Wang et al. [21]. The uncertainty associated with ϕ for the data from Wang et al. is shown for $\phi = 1.3$ alone b.) Power exponent (α) derived from present experiments vs. ϕ . Symbols correspond to experimental data and the lines represent simulations using the MB-MB2D mechanism.

the model is found to predict the experimental laminar burning velocities satisfactorily.

In addition to the ability of the mechanism in mimicking the present experimental results, a significant improvement is observed in predicting the laminar burning velocities at $T_g = 403$ K (Wang et al. [21], see Fig. 14(a)) when compared to the predictions of the model before revisions (Fig. 2(b)). At $T_g = 403$ K, the model predicts the burning velocities fairly satisfactorily at fuel-lean and stoichiometric conditions. At $\phi > 1.2$, the model overpredicts the experimental data, and a maximum deviation of 6.67 cm/s is observed at $\phi = 1.5$. This deviation can be attributed to the higher uncertainties in a premixed counter-flow burner at fuel-rich conditions [60–62]. Thus, the model well predicts the experimental data [21] if the uncertainty in the equivalence ratio is considered.

Figure 15 shows the effect of revisions in predicting S_L at 318 K and 403 K (red and black lines respectively). The solid lines represent the simulated S_L before any revisions in the MB-MB2D mechanism. After revising the MB2D kinetics, major improvements are seen in the prediction of S_L at both 318 and 403 K (—, — in red and black colours in Fig. 15). The dash-dotted line (— · — in red and black colours in Fig. 15) at 318 and 403 K represents the simulated S_L , before adding the new reactions involving O atom addition across the double bond in MB2D. After the revisions are introduced in MB2D+O chemistry as discussed in Section 3.2.5 [49], im-

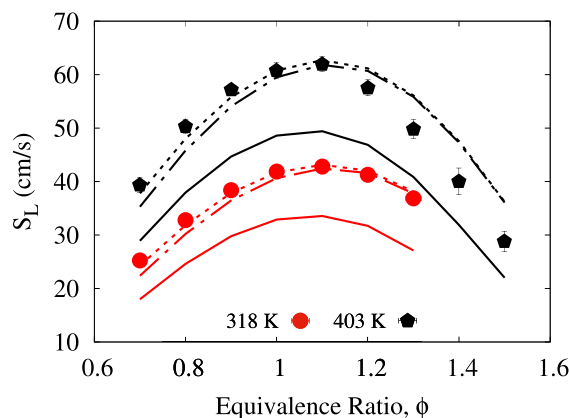


Fig. 15. Effect of revisions in MB2D chemistry on predicting laminar burning velocities (S_L). Symbols: Experimental data (current study and Wang et al. [21]). Solid lines: Simulated results before revisions. — · —: Simulated results after revisions. — · —: Simulated results after revisions and in the absence of new O-addition pathways.

provements are observed in the prediction of S_L at fuel-lean conditions. Thus, the revisions incorporated in the addition of O atom across the double bond [49] in MB2D are found to improve the reactivity of MB2D at lean fuel conditions, which is expected due the abundance of O atoms at lean conditions. At fuel-rich conditions, the updates in MB2D+O chemistry exhibit negligible effect in predicting S_L .

4.2.2. Burner-stabilized flame

Yang et al. [22] investigated the oxidation of MB2D/O₂/Ar mixtures at rich-fuel conditions ($\phi = 1.56$) by stabilizing the premixed flat-flame on a stainless steel McKenna burner at sub-atmospheric conditions (2.67 kPa). The species concentration measurements were reported to have an uncertainty of 20% for the major species and as large as a factor of 3 for other flame intermediates [22]. The study noted that the amounts of unsaturated intermediates in MB2D flames were higher when compared to the corresponding results of a saturated ester (MB). In order to understand the ability of the newly developed mechanism in predicting the species concentrations, the MB-MB2D mechanism has been employed to simulate the experimental premixed flame profiles. The simulations are done by imposing the experimental temperature profile in calculations and solving species transport equations.

Figure 16 shows the profiles of major and minor (C_2 and C_3) species in MB2D flame [22], along with the predictions of MB-MB2D mechanism. The simulated results of other minor species are provided in supplementary material (see Fig. S4). The simulated species profiles are shifted by 1 mm away from the burner, so as to account for the uncertainty in the location of measuring probe [22]. A very good agreement is observed between the simulated results and experimental species profiles for major fuel intermediates (Fig. 16(a and b)). In addition to the major species, the model predicts the peak concentration of various minor species (Fig. 16(c and d)) within the uncertainty levels (factor of 2).

The model is able to predict the peak values of C_2 species within the experimental uncertainties (Fig. 16(c)). However, the model overpredicts the amount of C_2 species after the peak (see Fig. 16(c)). This can be ascribed to the uncertainties in the photoionization cross sections for H₂O or C₂H₂ as reported by Yang et al. [22]. Besides, the overprediction of CH₃ radical (see Fig. S4(b)) increases the production of C₂H₆ by means of a CH₃–CH₃ recombination. The increased production of C₂H₆ can further contribute to the production of other C_2 species (C₂H₂ and C₂H₄). The overprediction of CH₃ radical can be attributed to the unaddressed

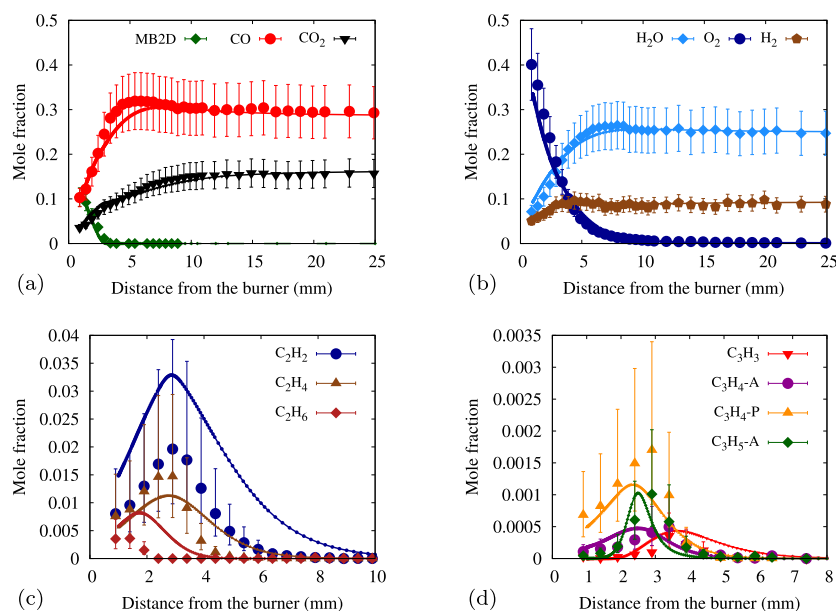


Fig. 16. Experimental and simulated species profiles for the oxidation of MB2D/O₂/Ar mixtures in a McKenna burner at $P = 2.67$ kPa and $\phi = 1.56$. Symbols correspond to experimental data from Yang et al. [22] and the lines with symbols represent the simulated profiles by using the MB-MB2D mechanism.

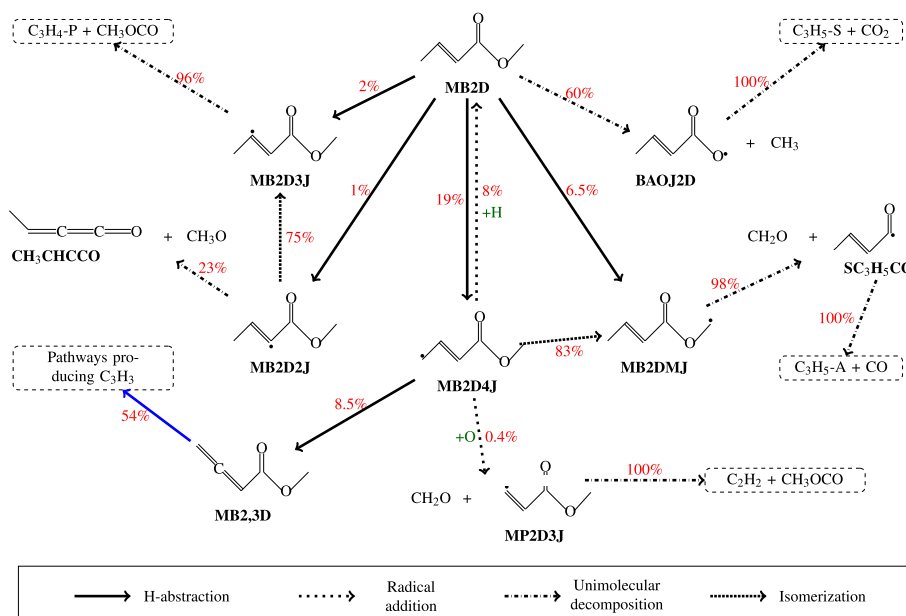


Fig. 17. Path flux analysis for the oxidation of MB2D/O₂/Ar mixtures at $\phi = 1.56$ and $P = 2.67$ kPa in a McKenna burner using the MB-MB2D mechanism.

pressure dependency of the unimolecular decomposition involving the O–CH₃ bond cleavage.

The revisions incorporated in the MB-MB2D mechanism is found to cause definite improvements in the prediction of various C₃ species (C₃H₄-A, C₃H₄-P, and C₃H₅-A) (see Fig. 16(d)) when compared to the predictions before revisions (see Fig. 2(a)). To understand the effect of various revisions on the observed improvements, a path flux analysis (Fig. 17) has been performed for the consumption of MB2D in McKenna burner. About 19% of MB2D is consumed through H-abstraction reaction producing MB2D4J and 6.5% producing MB2DMJ. In the absence of isomerization reactions (as in the MB-MB2D mechanism before revisions), the MB2D4J would have entirely reacted via an O atom addition followed by decomposition producing C₂H₂. After the revisions are incorporated, a large part of MB2D4J (83%) isomerizes to MB2DMJ. The

MB2DMJ radical further produces C₃H₅-A radicals (98%). In addition to the increased production of C₃H₅-A radicals through isomerization (MB2D4J → MB2DMJ), a portion of MB2D4J (8.5%) undergoes further H-abstraction (a new pathway introduced as part of the revisions), producing MB2,3D. This species, through various pathways (54%) boosts the production of C₃H₃. Also, the MB2D2J and MB2D3J radicals produced in minor quantities (1% and 2% respectively) contribute to the production of C₃ species like C₃H₄-P and CH₃CHCCO.

Thus, the revisions incorporated in the MB-MB2D mechanism boosts the production of various C₃ species like C₃H₃, C₃H₄-P, and C₃H₅-A, which can in turn contribute to the production of C₃H₄-A. The predictions of different major species are found to be good, while the C₂ species offers scope for further improvement in a future study. Overall, the ability of MB-MB2D mechanism in predict-

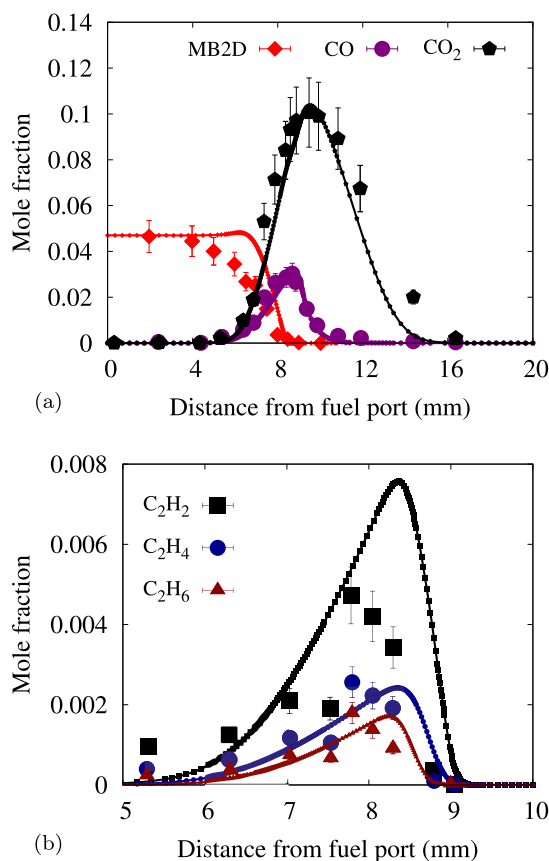


Fig. 18. Experimental and simulated species profiles for the oxidation of MB2D in a counter-flow diffusion flame burner at atmospheric conditions. Symbols: experimental data from Sarathy et al. [17] Lines with symbols: simulated profiles using the MB-MB2D mechanism.

ing various species is found to be improved, after the revisions are incorporated.

4.2.3. Counter-flow diffusion flame

In order to further ascertain the ability of the model in predicting MB2D combustion, the MB-MB2D mechanism has been employed to simulate the species profiles measured in counter-flow diffusion flame experiments at atmospheric conditions from the literature [17]. The experimental data, along with the simulated species profiles are presented in Fig. 18. The profiles of other minor species are included in the supplementary material (see Fig. S5). All experimental profiles are reported to have an uncertainty of 15% [17].

The model is found to predict the major species (MB2D, CO, and CO₂) satisfactorily as depicted in Fig. 18(a). Moreover, the model is found to successfully capture the experimental trends of other minor species such as CH₄, CH₂O, C₂H₂, C₂H₄, and C₂H₆ (see Fig. 18(b), Fig. S5 (a)). The species profiles of various minor species are presented in Fig. 18(b) and in the supplementary material (Fig. S5(b)). The MB-MB2D mechanism overpredicts the peak concentration of C₂H₂ by about 50%. This observation is consistent with the simulated results of the premixed MB2D flames (Fig. 16(c)). In addition to the above results, the model is found to capture the experimental trends of the C₃ intermediates (propane and propyne) accurately (see Fig. S5(b)). These results, along with the simulated results of the premixed MB2D flames [21] (Fig. 16), bolsters the ability of the MB-MB2D mechanism in predicting the species concentrations of MB2D oxidation.

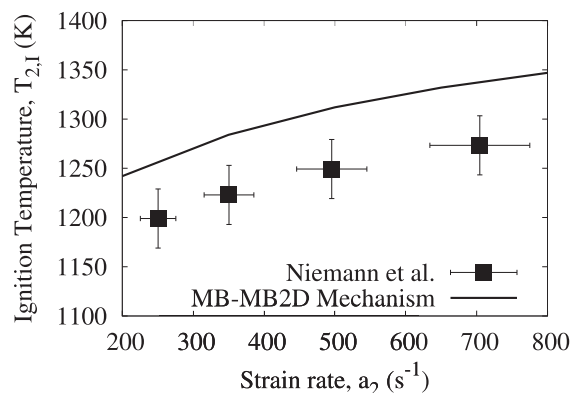


Fig. 19. Experimental and simulated results of the autoignition temperatures ($T_{2,I}$) of non-premixed MB2D flames. Symbols correspond to experimental data from Niemann et al. [23] and the lines represent simulated results using the MB-MB2D mechanism.

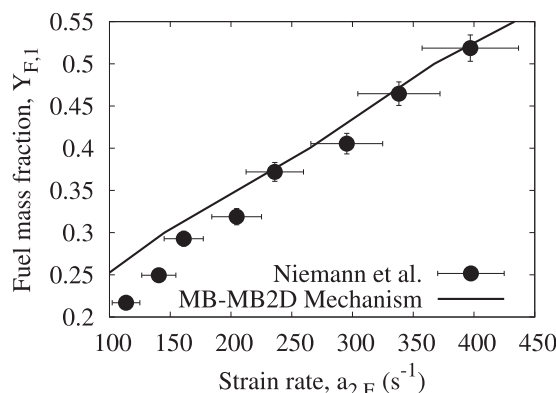


Fig. 20. Experimental and simulated results of the extinction strain rates ($a_{2,E}$) of non-premixed MB2D flames. Symbols correspond to experimental data from Niemann et al. [23] and the lines represent simulated results using the MB-MB2D mechanism.

4.3. Auto-ignition and extinction measurements

Critical conditions of autoignition and extinction in a non-premixed flame provide valuable additional experimental data to validate a mechanism. Besides, at critical conditions, the transition between a flame and no-flame solution depends on the competition between reaction kinetics and transport phenomena. Thus, the data obtained at critical conditions aid in evaluating the accuracy of the kinetic and transport model in the mechanism. Niemann et al. [23] performed experiments at critical conditions using MB2D and other ester fuels to obtain autoignition temperature ($T_{2,I}$) and extinction strain rates ($a_{2,E}$). The experiments were performed for non-premixed conditions in a counter-flow diffusion flame burner. The autoignition simulations are performed using FlameMaster [57], while extinction studies using Chemkin Pro [39].

Figure 19 shows the experimental autoignition temperatures ($T_{2,I}$) of the non-premixed MB2D flames [23], along with the simulated results. The momentum of the opposing axis-symmetric fuel (subscript 1) and oxidiser streams (subscript 2) are balanced ($\rho_1 V_1^2 = \rho_2 V_2^2$) at different oxidiser strain rates (a_2), for a fuel mass fraction ($Y_{F,1}$) of 0.4. The diffusion equation includes thermal diffusion and multi-geometric transport model, and the energy equation accounts radiative heat loss. A temperature shift of about 50–60 K is observed in the predictions of $T_{2,I}$ by the MB-MB2D mechanism. Nevertheless, the simulated results are found to follow the experimental trends of increasing $T_{2,I}$ as the strain rate increases.

Figure 20 presents the experimental extinction strain rates ($a_{2,E}$) vs. the fuel mass fraction ($Y_{F,1}$) of the non-premixed MB2D flames [23] along with the simulated results. The simulated results are found to emulate the experimental trend very well. The maximum deviation in $a_{2,E}$ is observed to be 47.9 s^{-1} , at $Y_{F,1} = 0.217$. Nonetheless, the ability of the mechanism in predicting the critical conditions of autoignition and extinction is found to be consistent and satisfactory.

5. Conclusion

A detailed kinetic mechanism (MB-MB2D mechanism) for the oxidation of methyl crotonate (MB2D) has been developed based on an existing MB2D model [18] (Gaïl et al., 2008), and a suitable base model [16] (Lele et al., 2018) in the literature. Major revisions have been introduced in the high-temperature kinetics of MB2D including the unimolecular decompositions, H-abstractions, and radical addition reactions. In addition to these revisions, the low-temperature chemistry of MB2D, involving successive addition of O_2 on the fuel radical has been included from our recent study [32]. The revisions were incorporated in the mechanism comprehensively, so as to put forth a detailed combustion mechanism for MB2D, valid for a wide range of conditions.

The validity of MB2D chemistry in the MB-MB2D mechanism is then assessed against a number of experimental studies in literature. These include ignition delay times in a shock tube and rapid compression machine, laminar burning velocities, species concentration studies under premixed and non-premixed configurations, and the critical conditions of ignition and extinction. The MB-MB2D mechanism has been found to successfully reproduce the trends in these experiments with good accuracy. In addition to above, the current study presents a new set of data for the laminar burning velocities of MB2D/air mixtures at different unburnt fuel-air temperatures. The MB-MB2D mechanism, when employed in simulations, accurately captures the measured laminar burning velocities.

In summary, the current study presents (i) a detailed kinetic mechanism for the combustion of MB2D, along with a comprehensive validation for the accuracy of MB2D kinetics and (ii) new experimental data for laminar burning velocities of MB2D at different temperatures, extending the experimental database for model validations. Following the current improvements in the kinetics of MB2D, a detailed comparative study will be done comparing the combustion of saturated (MB) and unsaturated esters (MB2D) in a separate work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.111409](https://doi.org/10.1016/j.combustflame.2021.111409)

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